

Key Insights

Developers:

Software developers and system architects pointed out several practical difficulties in managing design diagrams in real-world projects. In fast-paced agile development, requirements change frequently through user stories, client feedback, and bug fixes, but design documentation rarely gets updated. This creates a common problem of architectural drift, where existing diagrams no longer match the actual codebase. Manually updating diagrams in visual tools takes up valuable engineering time and is often neglected. Developers also face challenges in keeping different diagrams consistent with each other, such as making sure that actors in Use Case diagrams align with sequence flows and class structures. In distributed teams, different engineers use different tools, creating fragmented documentation. Additionally, developers expressed concerns about privacy and security, noting that proprietary project requirements and internal specifications cannot be uploaded to unverified online tools that store user data.

To solve these problems, developers want a practical tool that integrates smoothly with standard development workflows. They want bidirectional updates, so that when a requirement document is modified, the tool prompts the user to update the corresponding diagrams after showing a clear preview of changes. Developers strongly prefer text-based diagram outputs, such as PlantUML or Mermaid, so that diagrams can be committed directly to Git repositories, reviewed in pull requests, and kept close to the source code. They also emphasized the need for clear traceability linking each use case to a specific requirement ID. Intelligent checking for missing or contradictory requirements before generating diagrams, along with strict data privacy guarantees such as immediate file deletion and no permanent storage on servers, were identified as must-have features for professional adoption.